

ASCII Plasma Animation in C

A 19-line terminal animation using sine waves and ASCII characters

Aim

To generate an animated plasma effect in the terminal by combining sine waves and mapping the result to ASCII characters of varying density.

Prerequisites

Software

- GCC (or any C compiler)
- A Unix-like terminal (Linux, macOS, or WSL on Windows) with ANSI escape code support
- Math library (libm), linked with -lm

Knowledge

- Basic C: loops, pointers, printf / putchar
- Trigonometric functions (sin, hypot)
- ANSI escapes: `\033[2J` clears the screen, `\033[H` moves the cursor home
- Headers used: `stdio.h`, `math.h`, `unistd.h`

Source Code (plasma.c)

```
#include <stdio.h>
#include <math.h>
#include <unistd.h>

int main(void) {
    const char *shade = " .:-+*#%@";
    printf("\033[2J"); // clear screen once
    for (double t = 0; ; t += 0.15) {
        printf("\033[H"); // move cursor to top-left
        for (int y = 0; y < 24; y++) {
            for (int x = 0; x < 78; x++) {
                double v = sin(x * 0.12 + t) + sin(y * 0.25 - t * 1.3)
                    + sin((x + y) * 0.1 + t * 0.7)
                    + sin(hypot(x - 39, (y - 12) * 2) * 0.15 - t);
                putchar(shade[(int)((v + 4) / 8 * 9.99)]);
            }
            putchar('\n');
        }
        usleep(50000);
    }
}
```

Procedure

1. Save the program as `plasma.c`.
2. Define the shading string " .:-+*#%@", ordered lightest to densest.
3. Clear the screen once using `\033[2J`.
4. Start an infinite loop with time variable `t`, increasing by 0.15 per frame.
5. At the start of each frame, move the cursor to top-left with `\033[H` so the frame overwrites the previous one without flicker.

- For every cell in the 78 x 24 grid, sum four sine waves: horizontal, vertical, diagonal and radial (distance from centre).
- Normalise the sum (range -4 to +4) to an index 0-9 using $(v + 4) / 8 * 9.99$.
- Print the matching character with `putchar`, and a newline after each row.
- Pause 50 ms with `usleep(50000)` to control the frame rate.
- Compile and run (see below). Press Ctrl+C to stop.

Build and Run

```
gcc plasma.c -o plasma -lm
./plasma
```

Wave Components

Wave	Expression
Horizontal	$\sin(x * 0.12 + t)$
Vertical	$\sin(y * 0.25 - t * 1.3)$
Diagonal	$\sin((x+y) * 0.1 + t * 0.7)$
Radial	$\sin(\text{dist_from_centre} * 0.15 - t)$

Customisation

What to change	Where
Animation speed	<code>t += 0.15</code> or <code>usleep(50000)</code>
Grid size	<code>y < 24, x < 78</code> (and centre 39, 12)
Character palette	the shade string
Wave frequencies	multipliers inside <code>sin(...)</code>

Troubleshooting

- undefined reference to sin:** add `-lm` at the end of the compile command.
- Garbled output on Windows:** use WSL or Windows Terminal; old `cmd.exe` may not support ANSI codes.
- Flickering or scrolling:** make the terminal window at least 78 x 24 characters.

Result

The terminal displays a smooth, continuously flowing pattern of light and dense characters. The overlapping waves create shifting blobs and ripples, with a circular pulse radiating from the centre. The animation runs at roughly 20 frames per second without visible flicker.

Conclusion

Complex, organic-looking animation can be produced from very little code by superimposing simple periodic functions. The program demonstrates ANSI terminal control, mapping numeric values to visual output, and driving animation with a time variable. This technique is the basis of the classic

"plasma" effect from the demoscene.